Course: Cloud Data Engineering Laboratory Course Code: CSE421

Practical 1

Aim: Create a cloud storage bucket and upload a dataset (CSV/JSON). Demonstrate basic operations such as upload, download, and file organization using Amazon S3/Azure Blob Storage.

Objectives:
- To understand cloud object storage concepts.
- Create storage bucket/container.
- Upload/download and organize files.

Outcomes: Students will create buckets, upload/download datasets and organize files.

Theory: Cloud object storage stores data as objects in buckets/containers for scalable storage.

Input: AWS/Azure account and CSV/JSON dataset.

Output:

Steps:

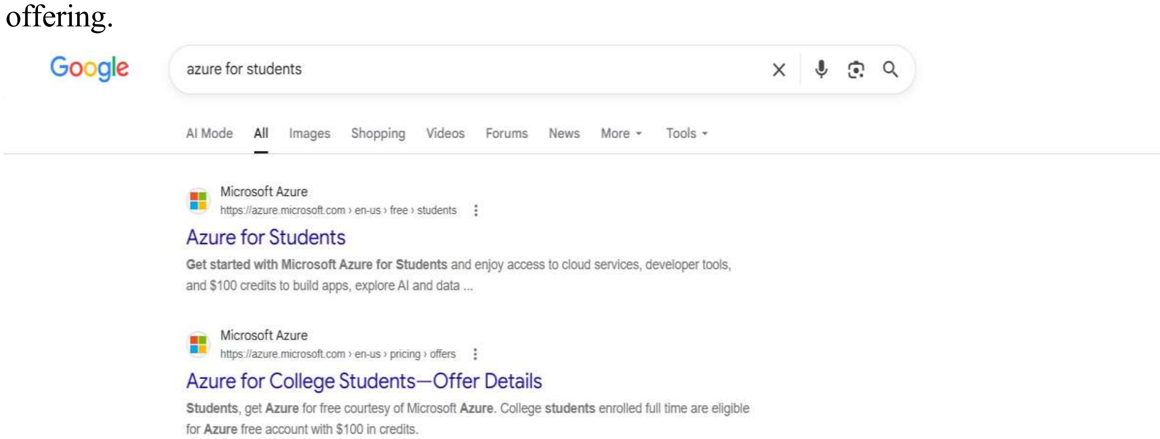
Account Creation, Resource Group, Storage Account, Container Setup & File Upload

This documentation serves as a comprehensive, step-by-step practical lab guide to configure Microsoft Azure cloud services. Follow these instructions to set up an Azure Student Account, create a Resource Group, deploy a Storage Account, create a Blob Container, and upload and verify files in the cloud.

1. Setting up an Azure for Students Account

Azure for Students provides $100 in free cloud credits and access to popular cloud services for 12 months without requiring a credit card.

- **Step 1.1:** Search for 'azure for students' on Google and select the official Microsoft Azure offering.

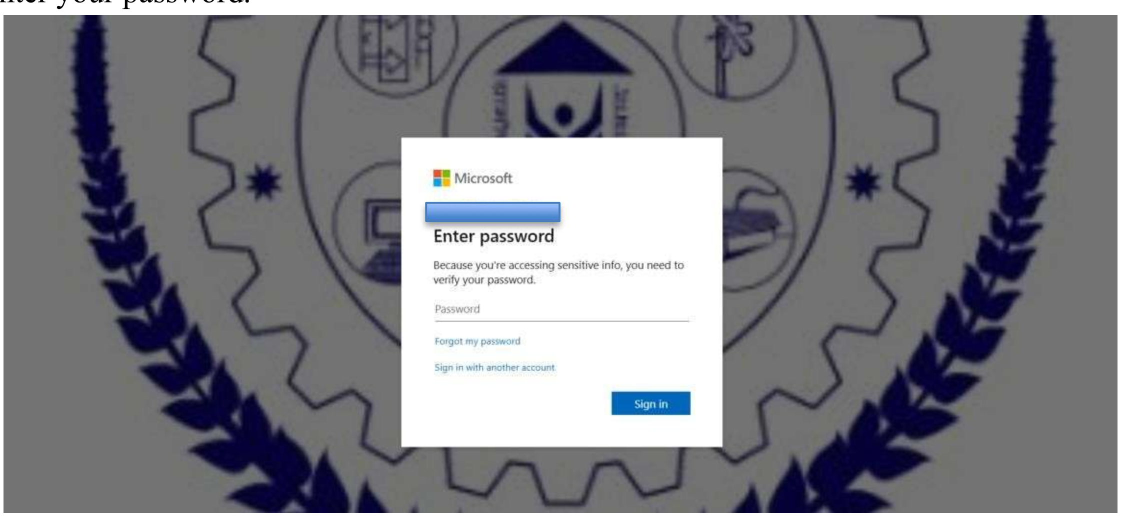


Step Image: Screenshot 2026-07-25 163106.png

- **Step 1.2:** Click on the blue 'Start free' button on the landing page.

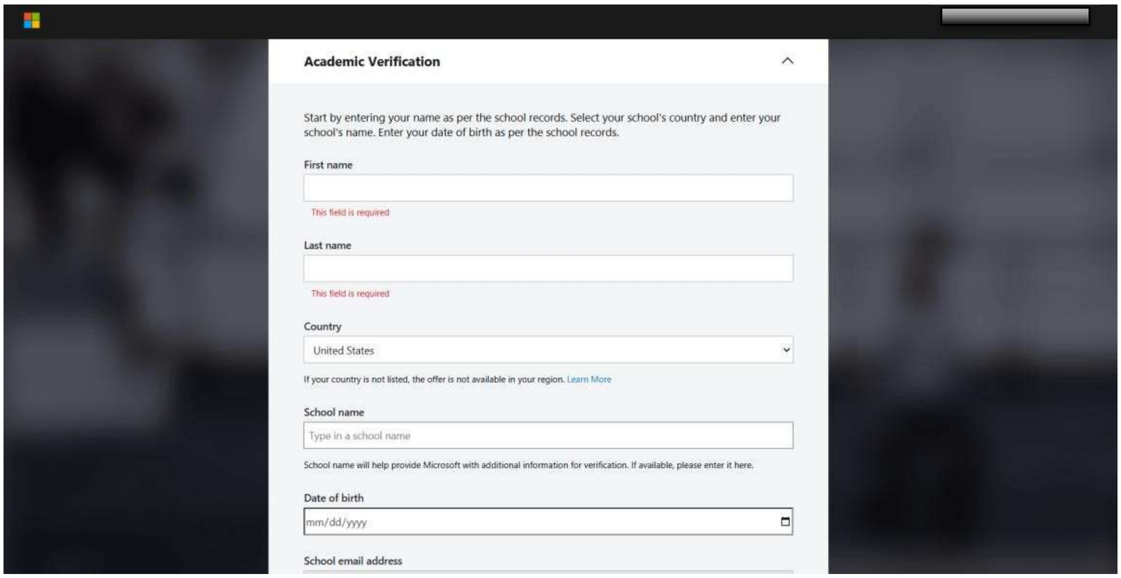
Step Image: Screenshot 2026-07-25 163136.png

- **Step 1.3:** Sign in with your academic organization email (e.g., yourname@mit.asia) and enter your password.



Step Image: Screenshot 2026-07-25 163224.png

- **Step 1.4:** Complete the Academic Verification form. Fill in details like your name, school, date of birth, and academic email.



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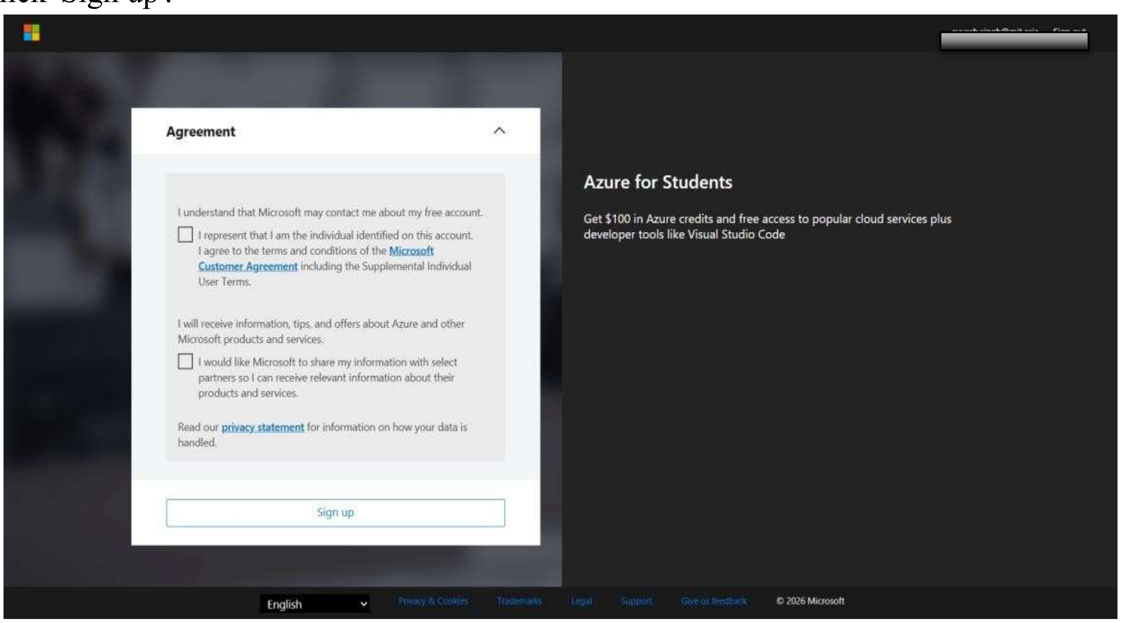
- **Step 1.5:** Review the Microsoft Customer Agreement, check the verification boxes, and click 'Sign up'.



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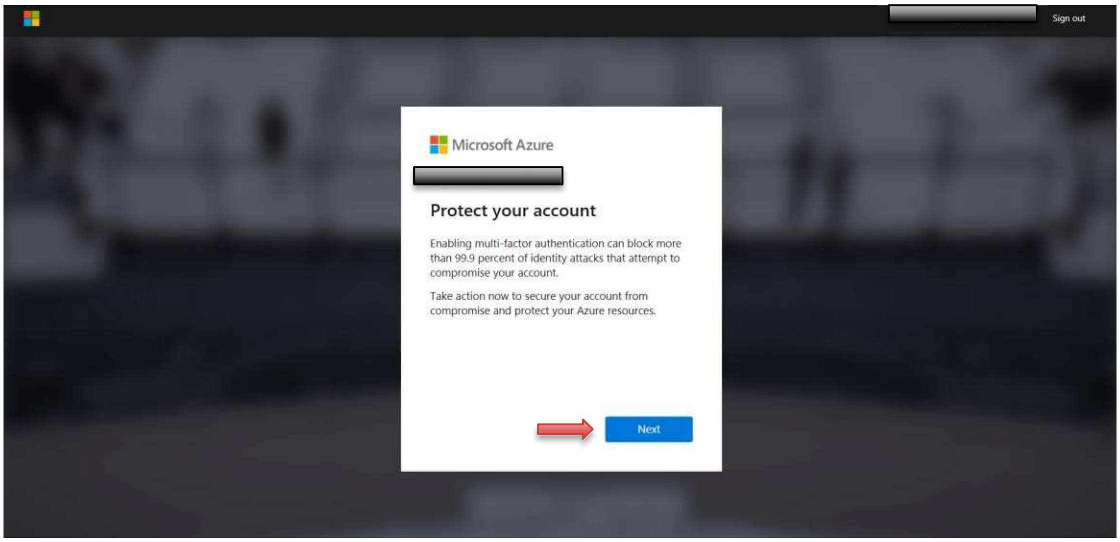
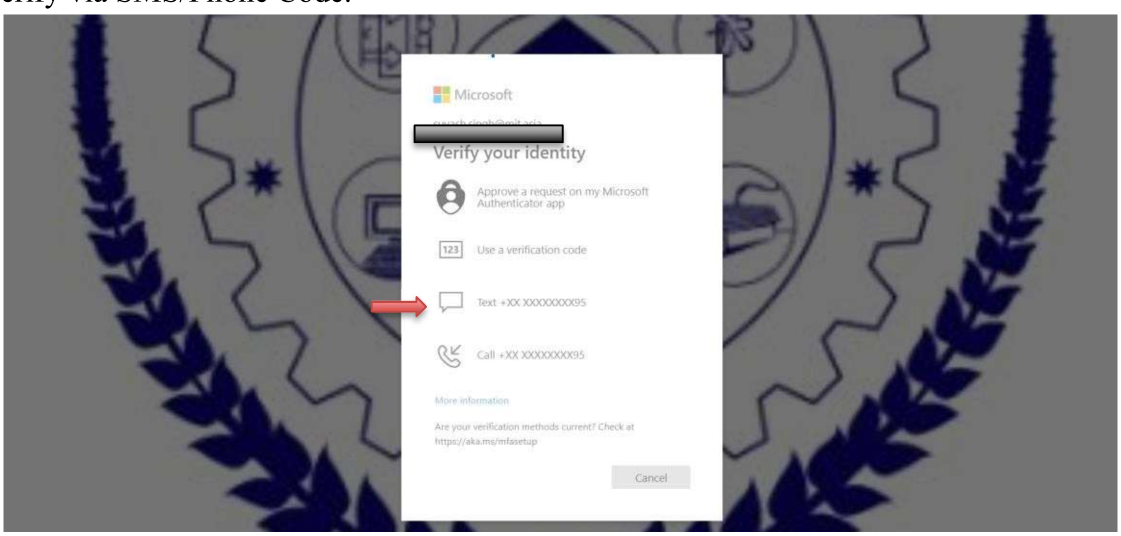
- **Step 1.6:** Set up Multi-Factor Authentication (MFA) to secure your account. Click 'Next' on the prompt.



Step Image: Screenshot 2026-07-25 163614.png

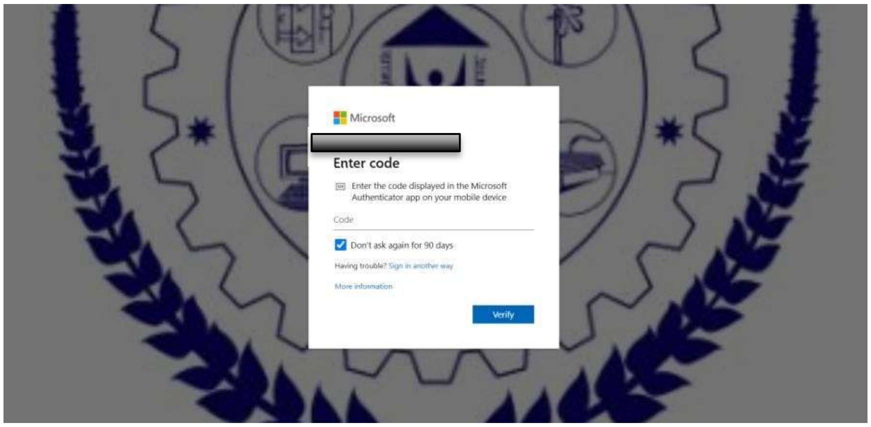
- **Step 1.7:** Choose a verification method: You can use the Microsoft Authenticator App or verify via SMS/Phone Code.



Step Image: Screenshot 2026-07-25 163817.png

- **Step 1.8:** If using Authenticator, enter the code from your app. If using SMS, enter the verification code texted to your registered phone number.

\- Authenticator Verification:



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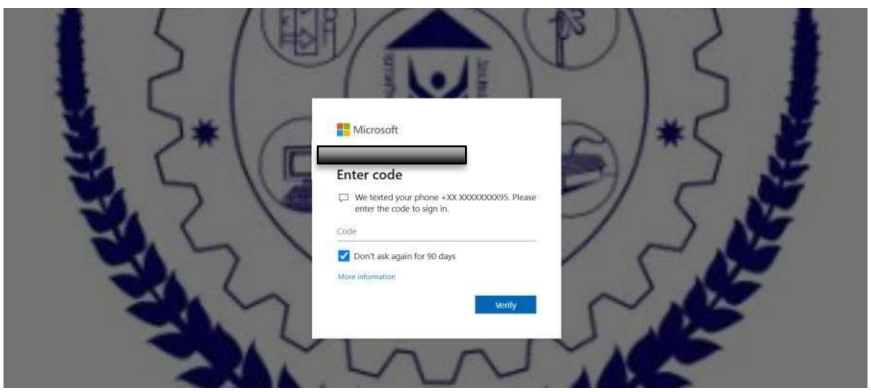
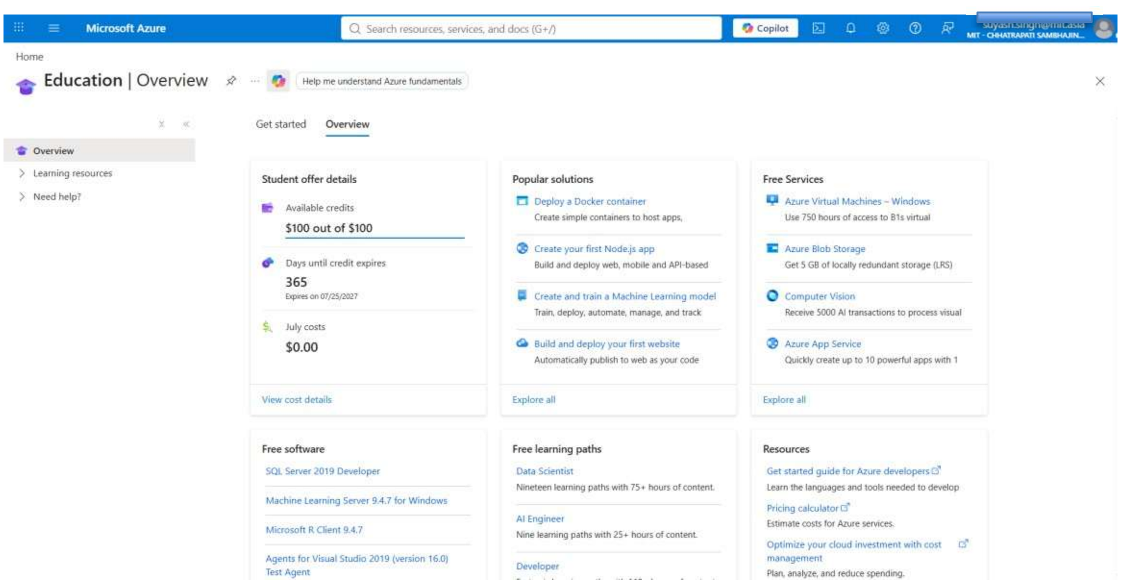
- Phone SMS Verification:



Step Image: Screenshot 2026-07-25 163829.png

- **Step 1.9:** Once identity verification is successful, the browser redirects you to the Azure Portal Education | Overview page. Verify that your $100 credits and 365 days of validity are active.

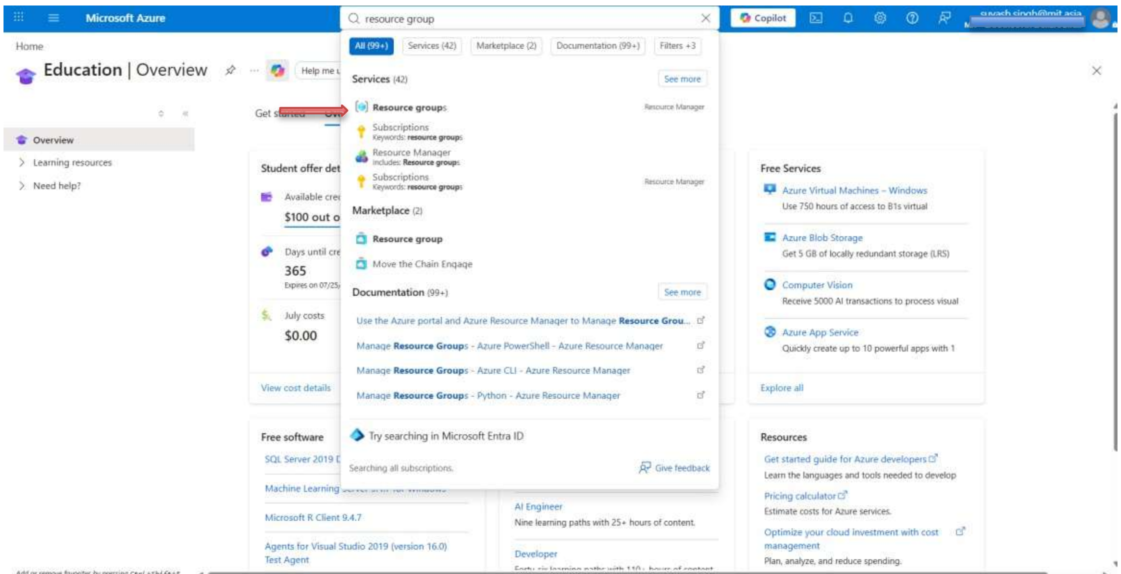


Step Image: Screenshot 2026-07-25 163915.png

2. Creating a Resource Group

A Resource Group serves as a logical metadata folder that groups your Azure resources for billing, security, and administrative management.

- **Step 2.1:** Type 'resource group' in the portal's top global search box and click on the 'Resource groups' service.



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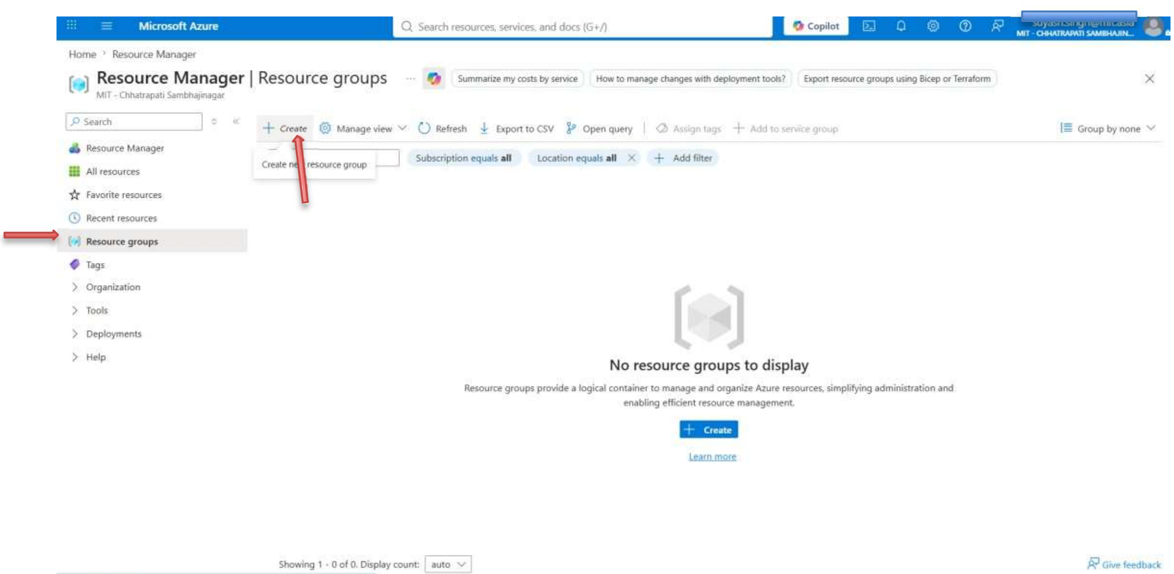
- **Step 2.2:** On the Resource Groups folder page, click the '+ Create' button on the toolbar.



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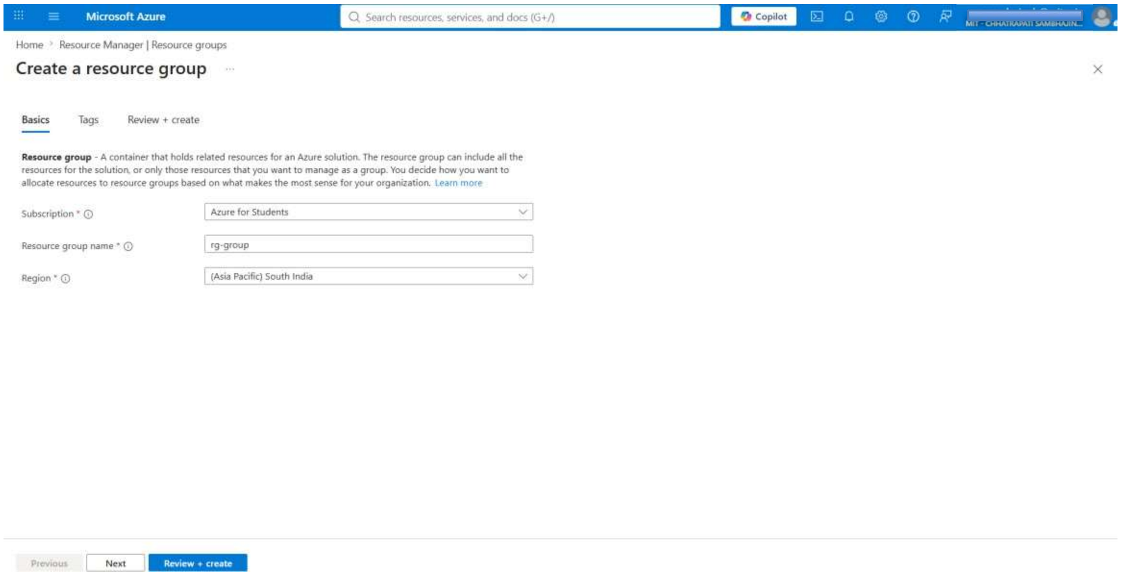
- **Step 2.3:** In the 'Basics' tab, set the Subscription to 'Azure for Students', name the Resource Group 'rg-group', and select a deployment Region (e.g., South India).



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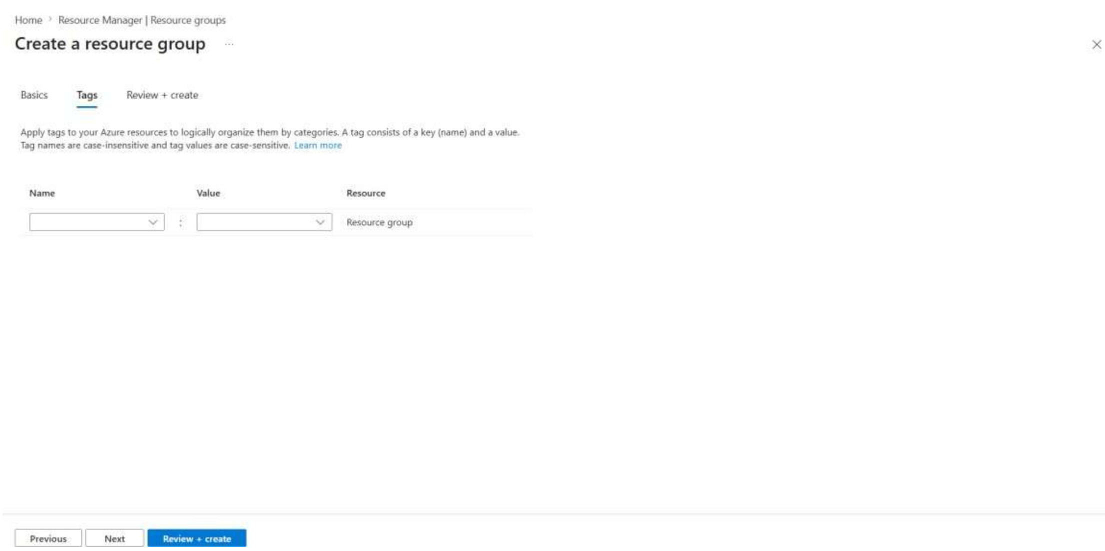
- **Step 2.4:** Click 'Next: Tags' to proceed (Tags are optional metadata; leave them empty).



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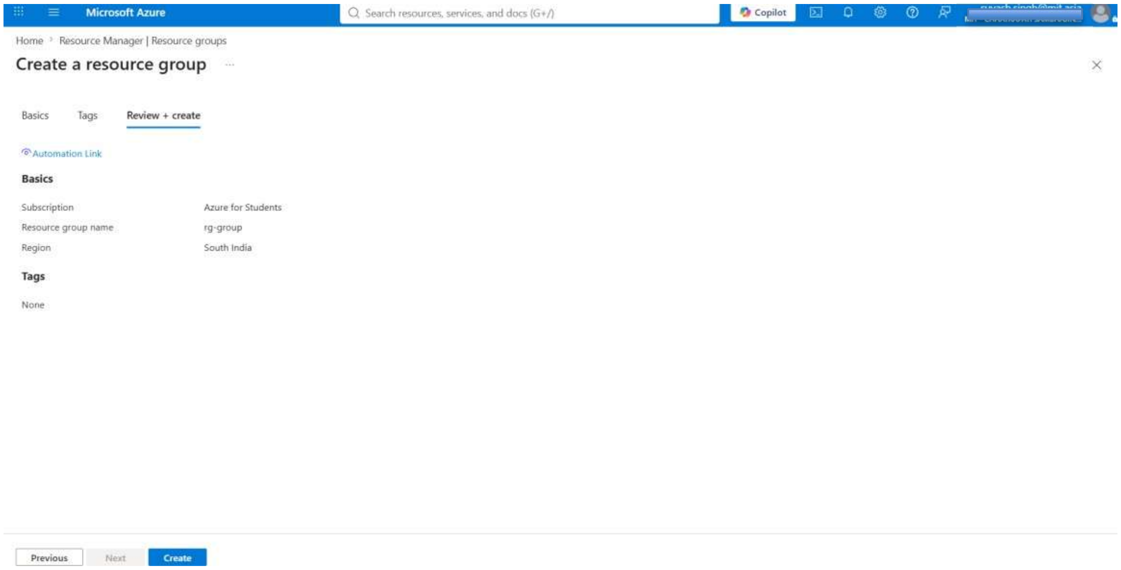
- **Step 2.5:** Click 'Review + create'. Once validation succeeds, verify the parameters and click 'Create'.



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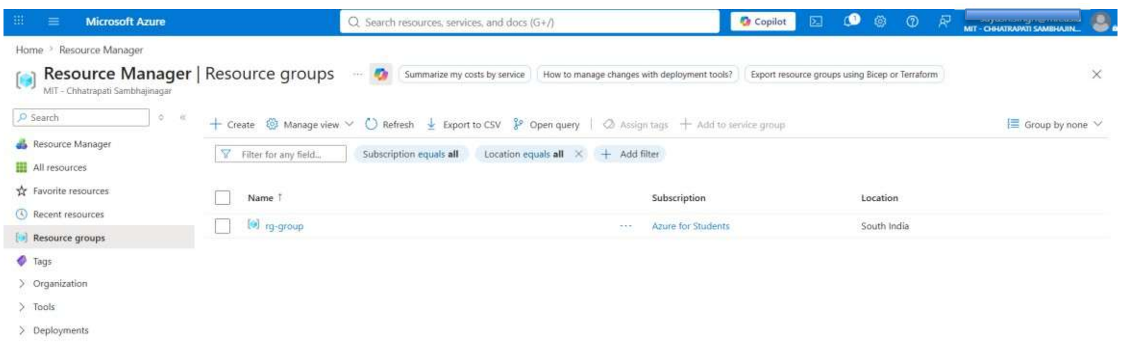
- **Step 2.6:** Confirm that your resource group 'rg-group' appears successfully in your list.

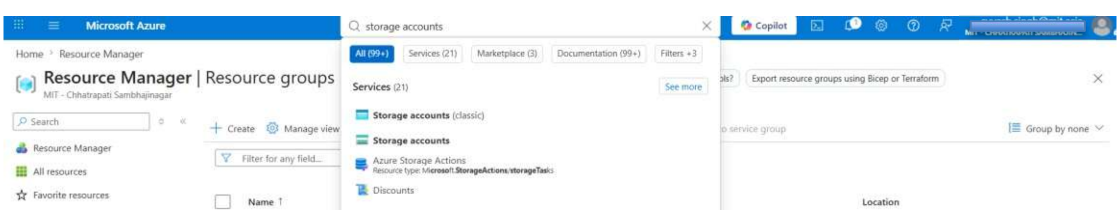


Step Image: Screenshot 2026-07-25 164247.png

3. Creating a Storage Account

An Azure Storage Account contains all of your cloud storage resources: blob containers, files, queues, and tables.

- **Step 3.1:** Search for 'storage accounts' in the top search bar and click on the 'Storage accounts' service.



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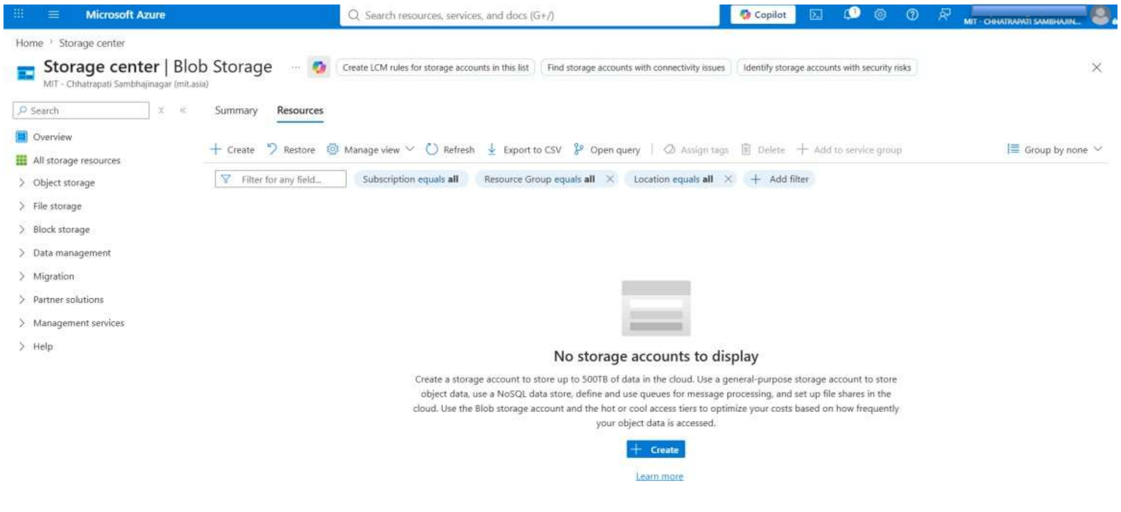
- **Step 3.2:** Click the '+ Create' button to deploy a new storage instance.



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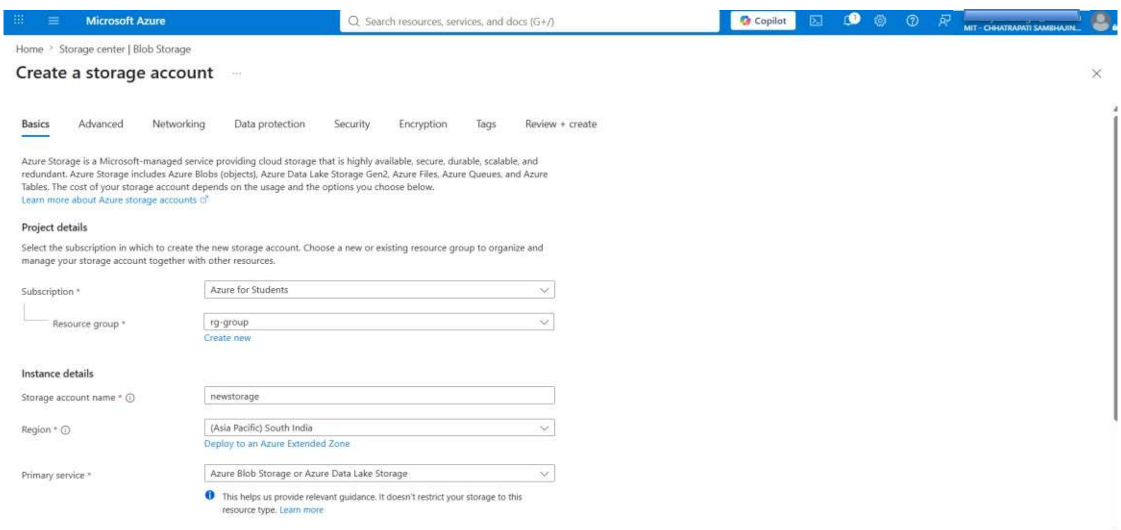
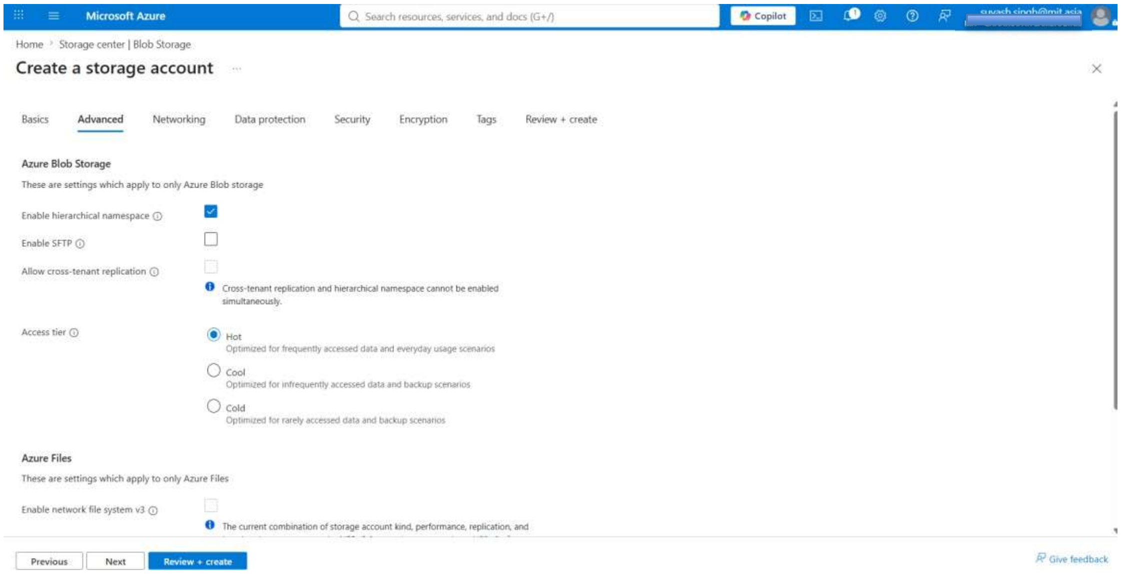
- **Step 3.3:** In the 'Basics' tab, select 'Azure for Students' subscription and the resource group 'rg-group'. Under 'Storage account name', type a unique name. Select a region, standard performance, and GRS or LRS redundancy.



Step Image: Screenshot 2026-07-25 164603.png

> **[Warning]** *Storage account names must be globally unique across all of Azure. If a common name like 'newstorage' is taken, try a unique name like 'mystorage879'. Also, you may need to adjust the deployment region based on quota availability.*

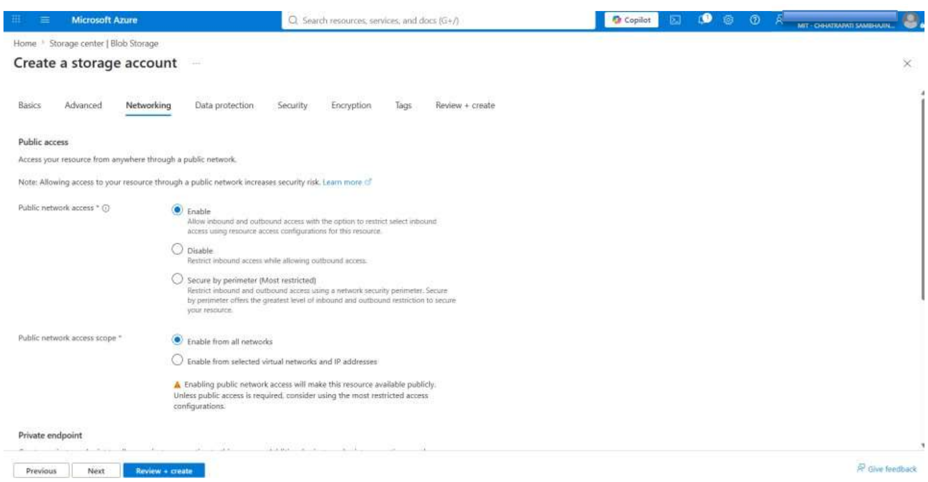
- **Step 3.4:** Under the 'Advanced' tab, ensure 'Enable hierarchical namespace' is checked. This enables ADLS Gen2, which is ideal for big data analytics operations.



Step Image: Screenshot 2026-07-25 164621.png

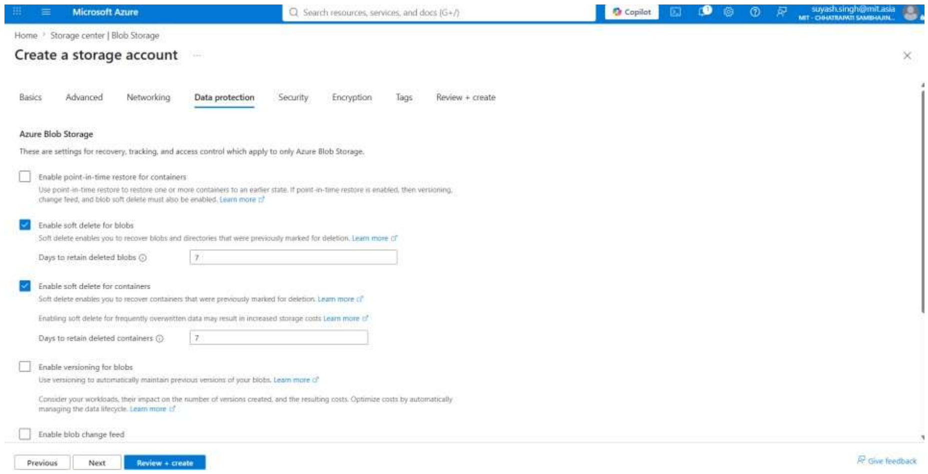
- **Step 3.5:** Review other configuration tabs like Networking, Data protection, Security, Encryption, and Tags. Keep their default parameters.

\- Networking Tab (Public access from all networks enabled):



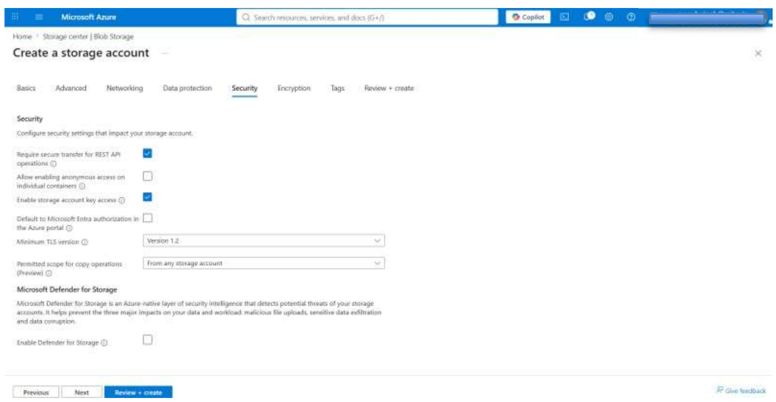
Step Image: Screenshot 2026-07-25 164634.png

\- Data Protection Tab (Soft delete enabled for safety):

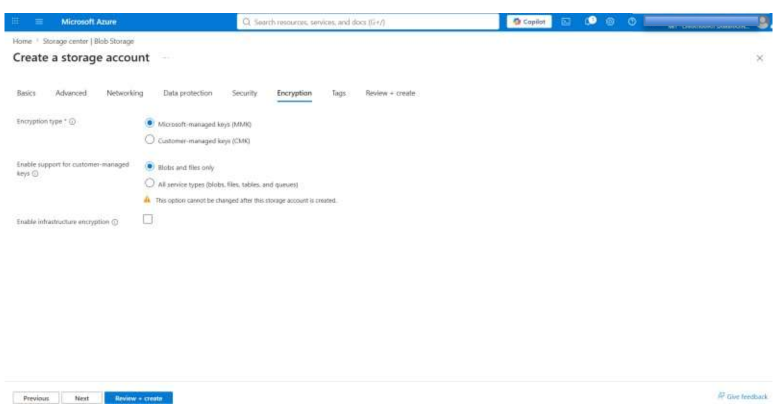


Step Image: Screenshot 2026-07-25 164649.png

- Security and Encryption Tabs:



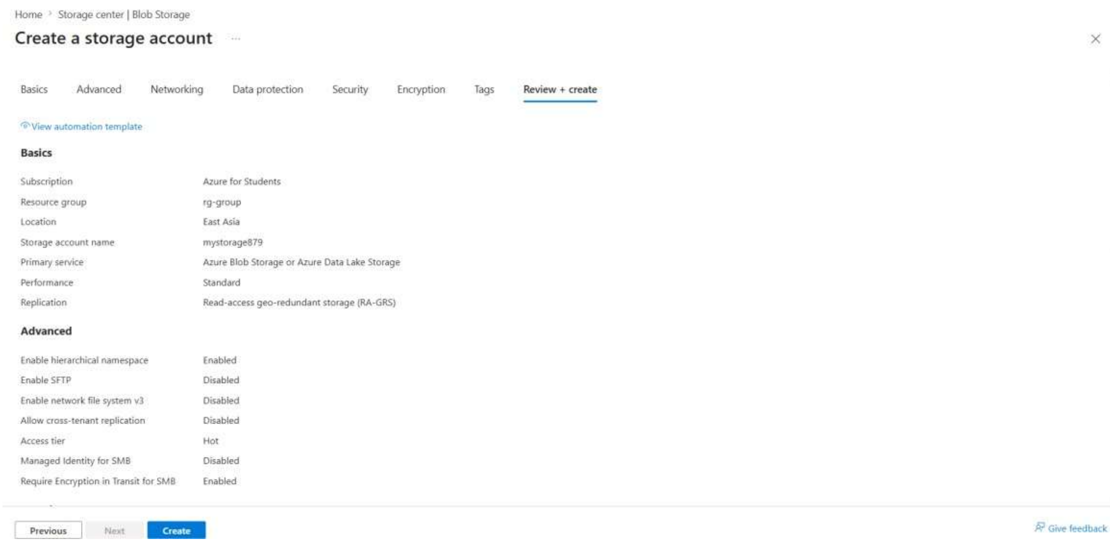
Step Image: Screenshot 2026-07-25 164703.png



Step Image: Screenshot 2026-07-25 164718.png

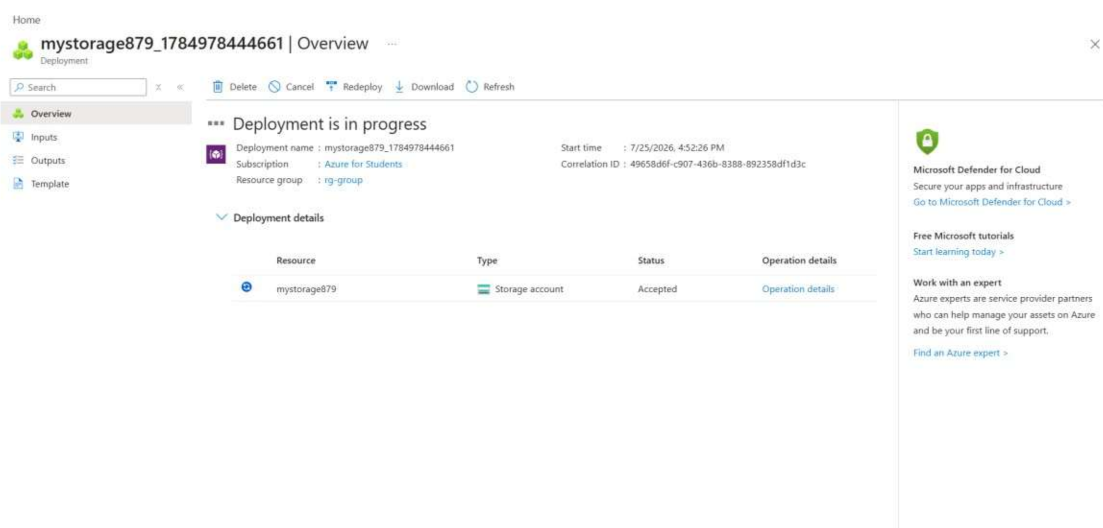
Step Image: Screenshot 2026-07-25 164751.png

- **Step 3.6:** In the 'Review + create' tab, click 'Create' after validation is successful. (Notice name changed to 'mystorage879' in region 'East Asia').



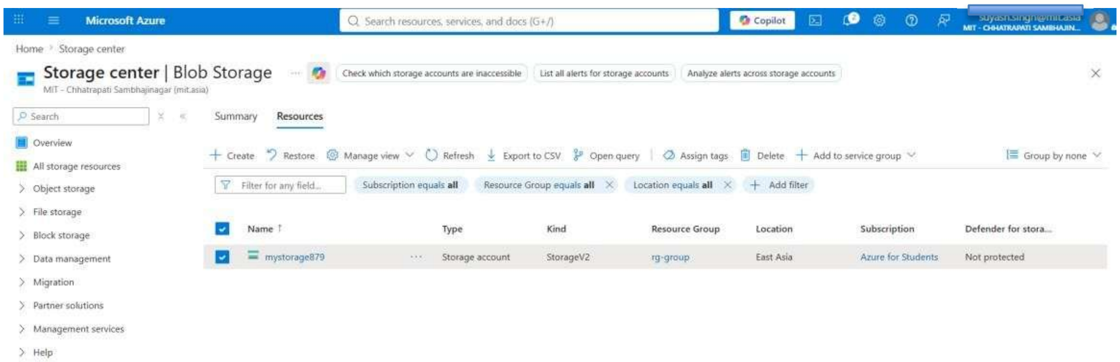
Step Image: Screenshot 2026-07-25 165158.png

- **Step 3.7:** Monitor the progress page which displays 'Deployment is in progress'.



Step Image: Screenshot 2026-07-25 165251.png

- **Step 3.8:** Once done, confirm that your storage account 'mystorage879' appears active in the storage accounts directory.

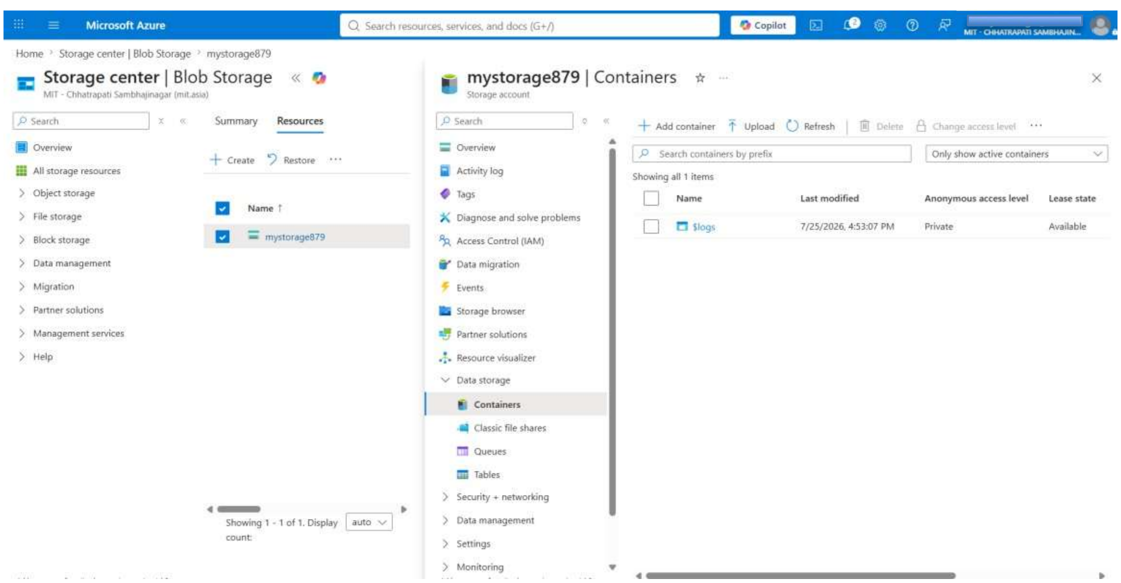


Step Image: Screenshot 2026-07-25 165436.png

4. Creating a Container

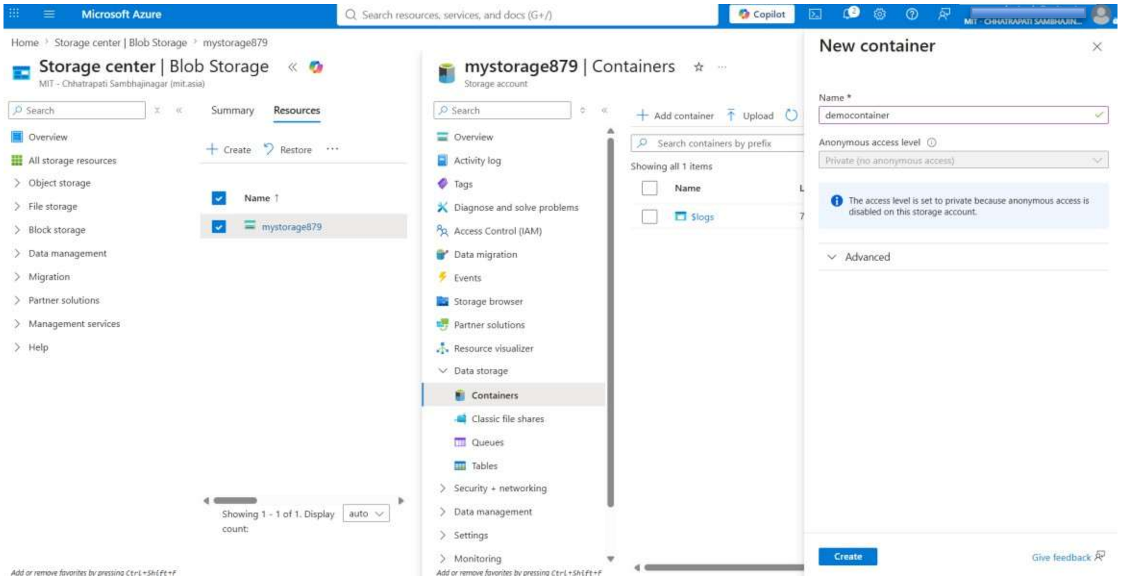
Azure Blob storage organizes data in containers, analogous to folders on a disk.

- **Step 4.1:** Open the 'mystorage879' storage account dashboard, find 'Containers' in the left menu under 'Data storage', and click '+ Add container'.



Step Image: Screenshot 2026-07-25 171346.png

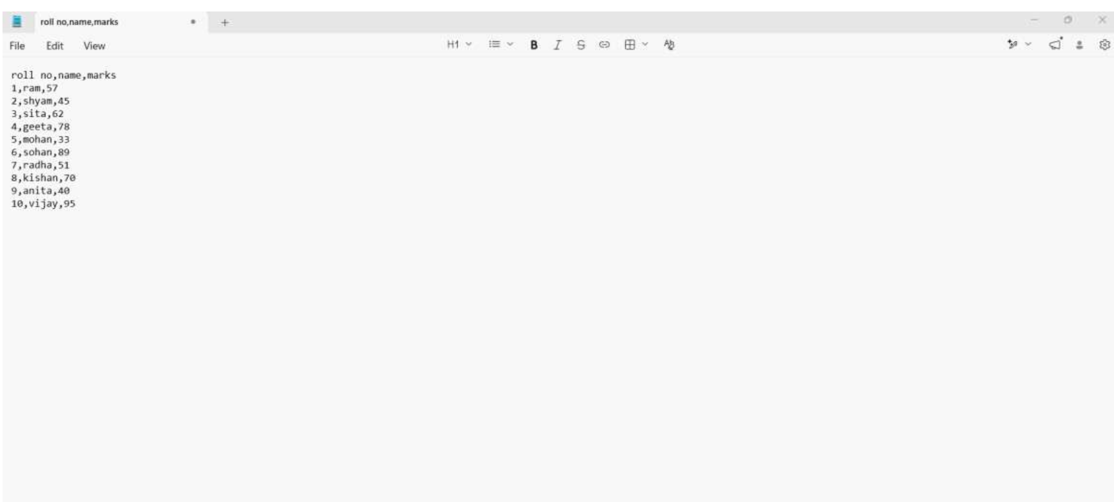
- **Step 4.2:** Name the container 'democontainer', leave the Access Level as 'Private', and click 'Create'.



Step Image: Screenshot 2026-07-25 171409.png

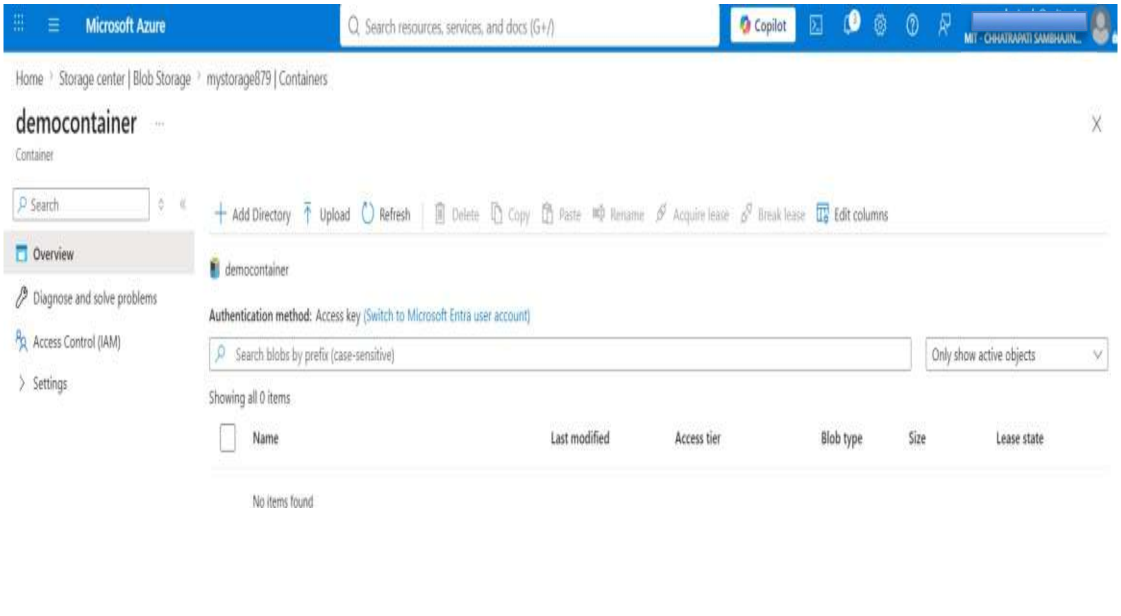
5. Preparing and Uploading a File

- **Step 5.1:** On your computer, open a text editor (Notepad) and write student data in CSV format. Save it locally as 'demo.csv'.



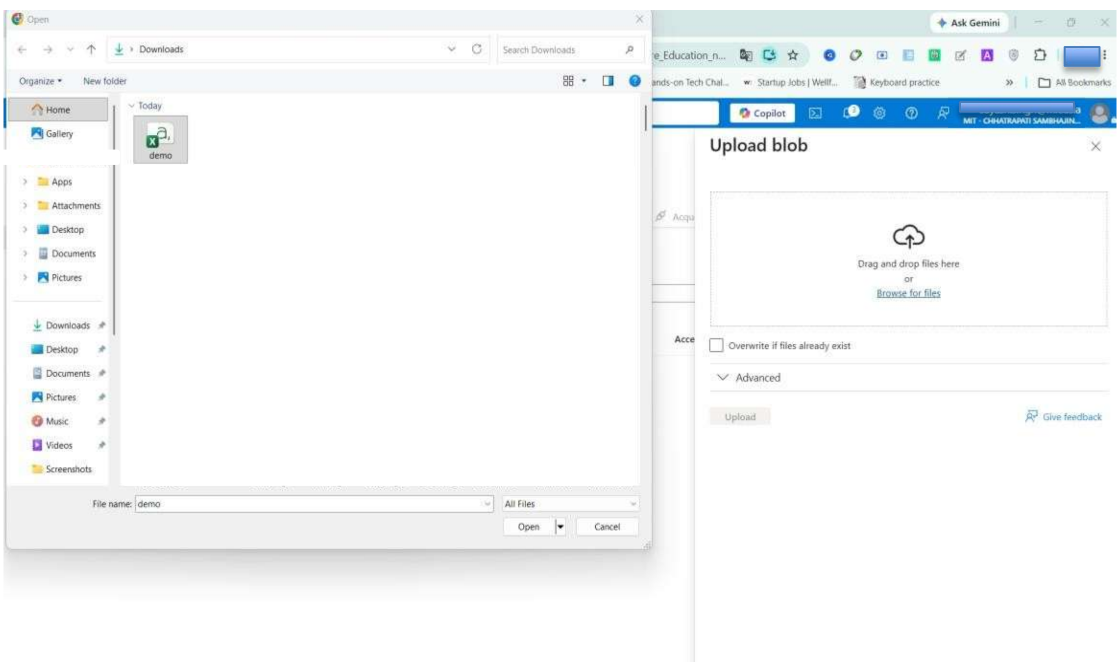
Step Image: Screenshot 2026-07-25 171117.png

- **Step 5.2:** Inside your container 'democontainer' in the portal, click the 'Upload' button on the toolbar.



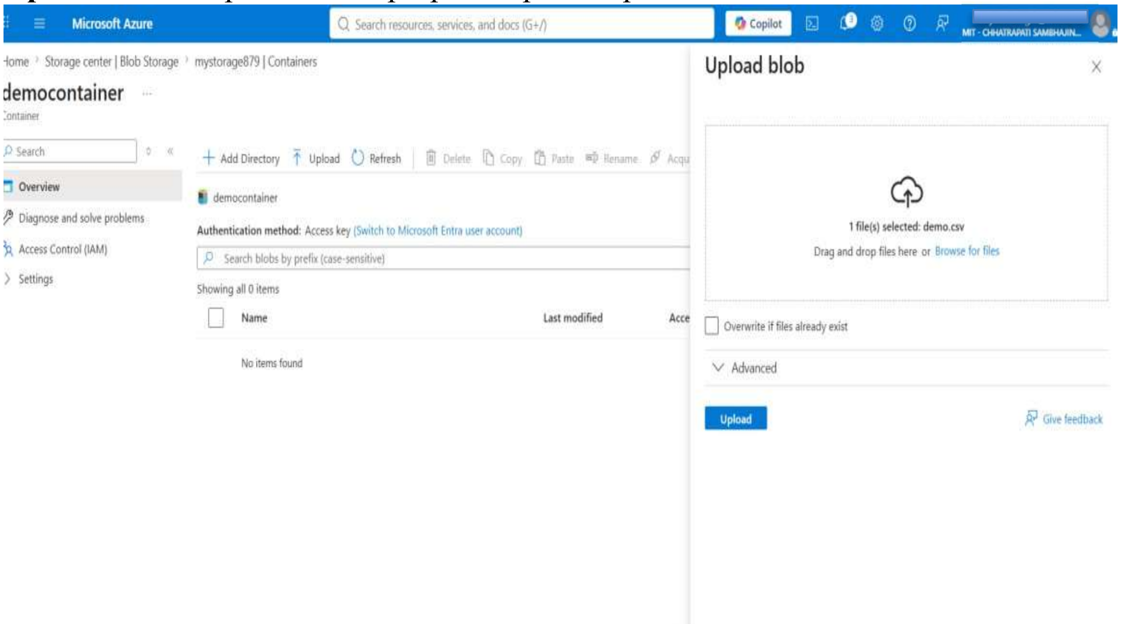
Step Image: Screenshot 2026-07-25 171427.png

- **Step 5.3:** In the panel, browse and select the 'demo.csv' file from your system downloads.



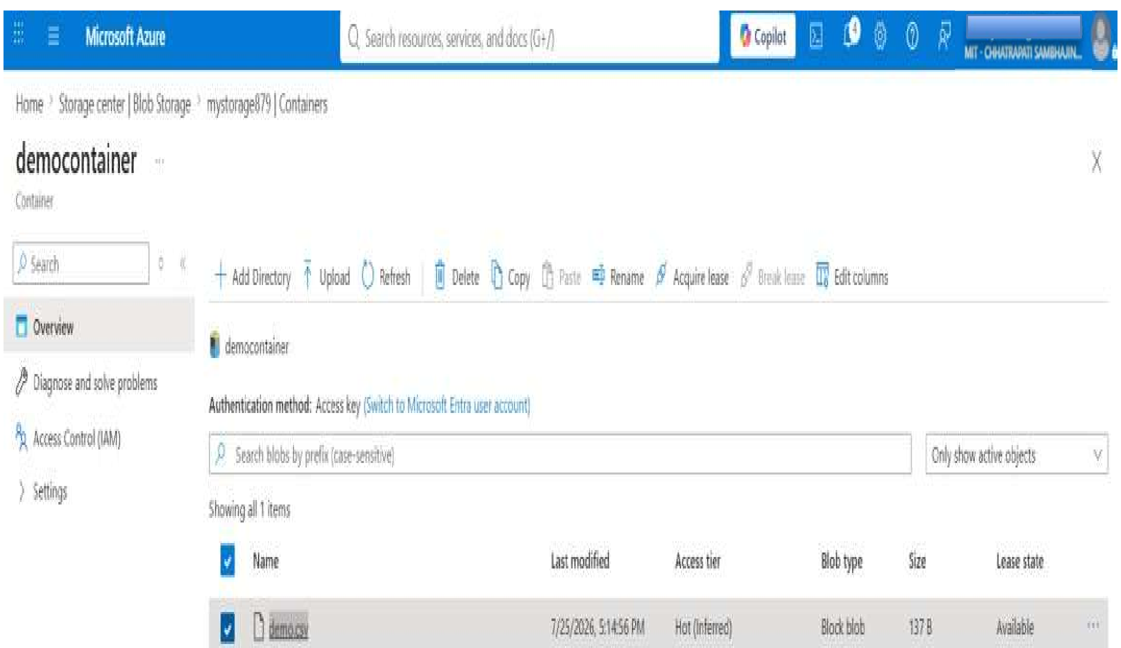
Step Image: Screenshot 2026-07-25 171442.png

- **Step 5.4:** Click 'Upload' in the properties pane to push the file to the cloud.



Step Image: Screenshot 2026-07-25 171453.png

- **Step 5.5:** Verify that 'demo.csv' is listed in the container's contents.



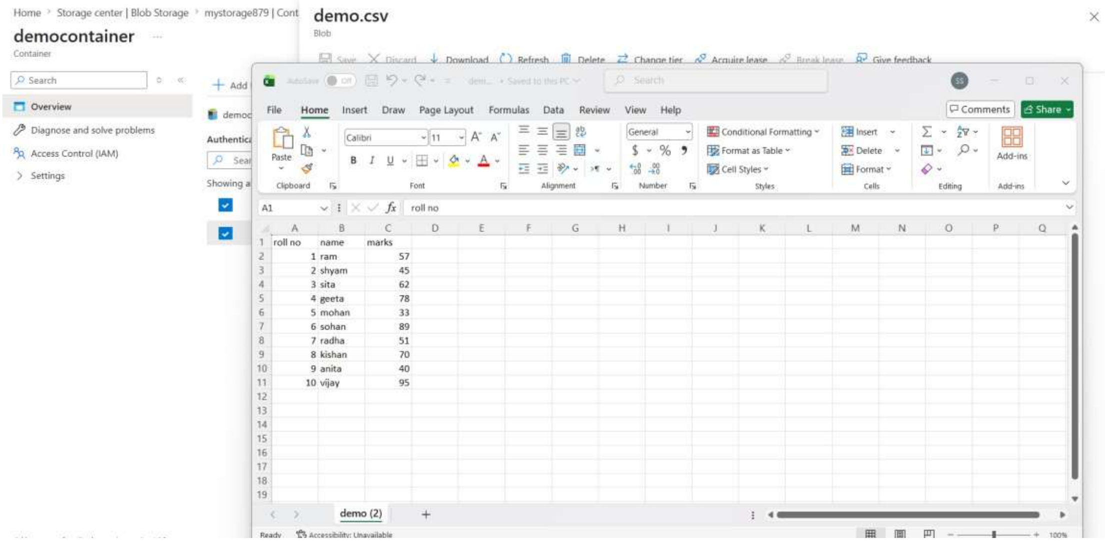
Step Image: Screenshot 2026-07-25 171552.png

6. Verifying the Uploaded File

You can verify the successfully uploaded data through multiple approaches:

Method 1: Local Application Verification

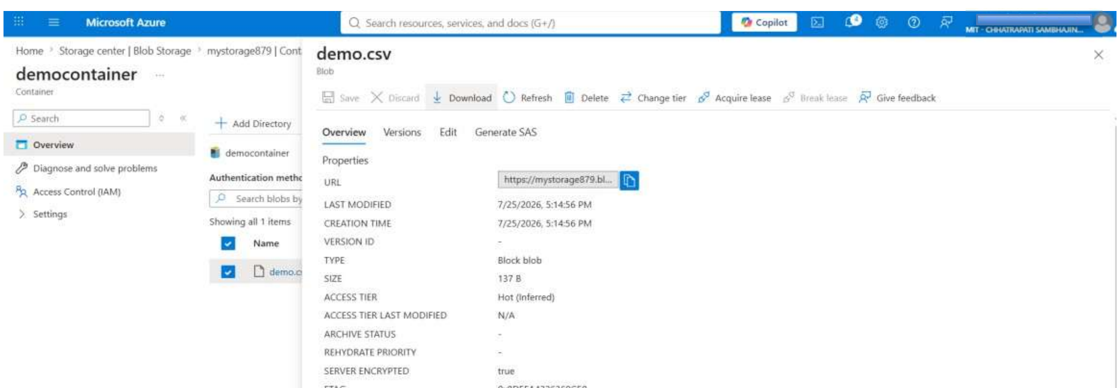
Open or double-click the original CSV file locally using Microsoft Excel to check column alignment (roll no, name, marks).



Step Image: Screenshot 2026-07-25 171636.png

Method 2: Metadata Inspection

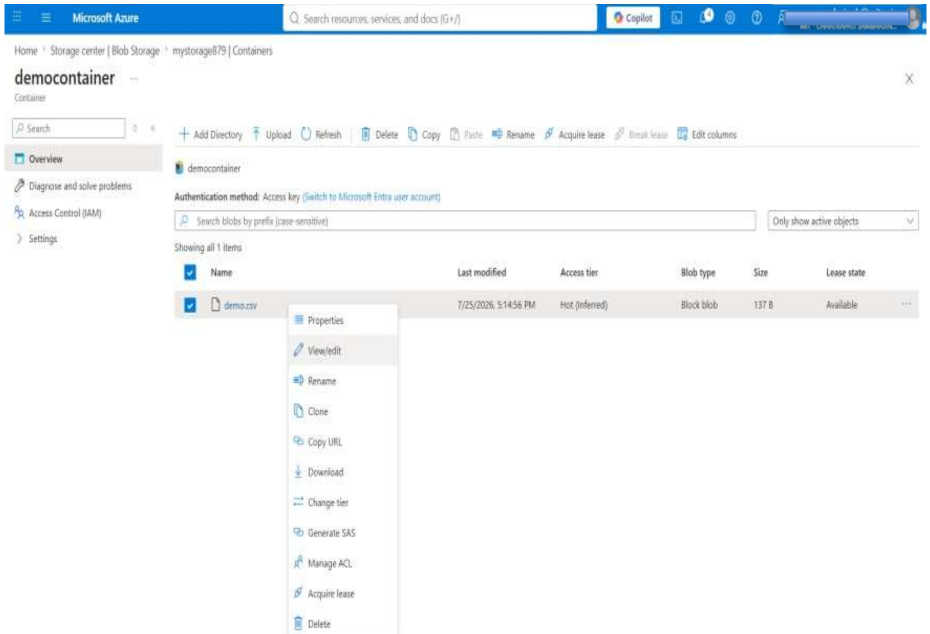
Click on 'demo.csv' in the container view. Under 'Overview', examine its properties: size (137 B), Blob type (Block blob), and copy the endpoint URL.



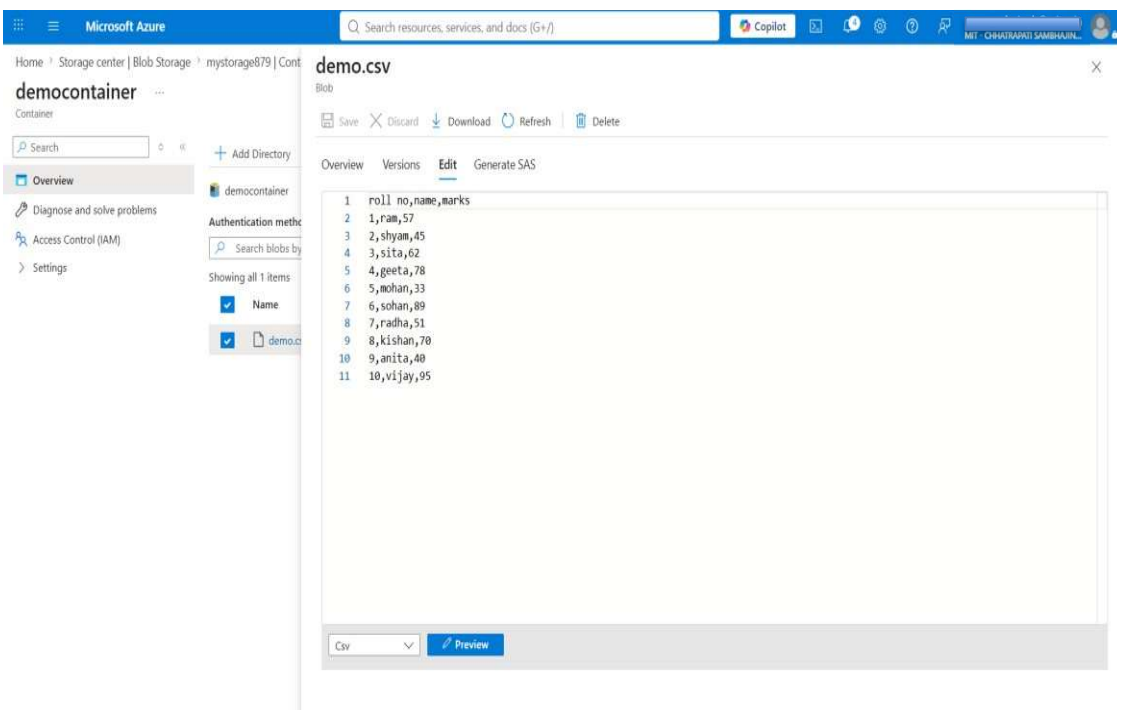
Step Image: Screenshot 2026-07-25 171612.png

Method 3: In-Portal Raw Text View

Click on the file options (...) and choose 'View/edit' to inspect the CSV text formatting directly in the cloud editor.



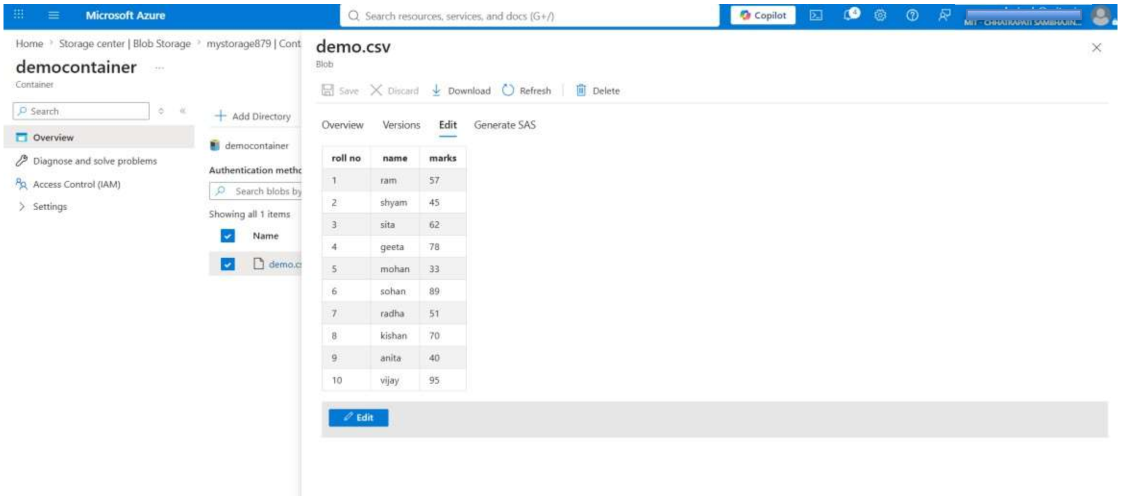
Step Image: Screenshot 2026-07-25 171658.png



Step Image: Screenshot 2026-07-25 171718.png

Method 4: Structured Cloud Preview

Click the 'Preview' button in the cloud editor. Azure parses the CSV and renders the student data in a clear grid/table.



Step Image: Screenshot 2026-07-25 171729.png

Practical Related Questions:

- What is object storage?
- Differentiate Amazon S3 and Azure Blob Storage.
- What is a storage bucket?

Conclusion: The practical was successfully completed by creating a cloud storage bucket/container and uploading a dataset in CSV/JSON format. Basic cloud storage operations such as uploading, downloading, and organizing files into folders were demonstrated using Azure Blob Storage. This practical provided hands-on understanding of how cloud storage enables scalable, reliable, and easily accessible data storage and management.